particular way, the rules of quantum mechanics can tell you how the wave function will change over time, and what the probability of various possible measurement outcomes will be should you choose to observe it.

The fact that the quantum recipe provides us with probabilities rather that certainties might be annoying, but we could learn to live with it. What bugs us, or should, is our lack of understanding about what is actually happening.

Imagine that some devious genius figured out all the laws of physics, but rather than revealing them to the rest of the world, they programmed a computer to answer questions concerning specific physics problems, and put an interface to the program on a web page. Anyone who was interested could just surf over to that site, type in a well-posed physics question, and get the correct answer.

Such a program would obviously be of great use to scientists and engineers. But having access to the site wouldn't qualify as understanding the laws of physics. We would have an oracle that was in the business of providing answers to specific questions, but we ourselves would be completely lacking in any intuitive idea of the underlying rules of the game. The rest of the world's scientists, presented with such an oracle, wouldn't be moved to declare victory; they would continue with their work of figuring out what the laws of nature actually were.

Quantum mechanics, in the form in which it is currently presented in physics textbooks, represents an oracle, not a true understanding. We can set up specific problems and answer them, but we can't honestly explain what's happening behind the scenes. What we do have are a number of good ideas about what that could be, and it's past time that the physics community started taking these ideas seriously.

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* There's a slight technicality, which we'll mention here and then pretty much forget about: the amplitude for any given outcome is actually a complex number, not a real number. Real numbers are the ones that appear on the number line, any number between minus infinity and plus infinity. Anytime you take the square of a real number, you get another real number that is greater than or equal to zero, so as far as real numbers are concerned there's no such thing as the square root of a negative number. Mathematicians long ago realized that square roots of negative numbers would be really useful things to have, so they defined the "imaginary unit" i as the square root of -1. An imaginary number is just a real number, called "the imaginary part," times i. Then a complex number is just a combination of a real number and an imaginary one. The little bars in the notation $|\text{Amplitude}|^2$ in the Born rule mean that we actually add the squares of the real and the imaginary